

COURSE TITLE: EMPIRICAL RESEARCH METHODS FOR COMPUTER SCIENTISTS			
Course Code:	22PCA802	Course Type:	CC
Teaching Hours/ Week (L:T:P):	4:0:0	Credits:	4
Total Teaching Hours:	60+0+0	CIE + SEE Marks:	50+50=100

Preamble (brief description).

Understanding the various empirical research methods, including quantitative and qualitative approaches, as well as their applicability, is essential for conducting rigorous and impactful research in computer science.

Course Outcomes

At the end of the course students will be able to...

CO1	Discuss the various empirical research methods for computer science and their applicability
CO2	Conduct quantitative research on computing research problems
CO3	Collect and analyse data after conducting experiments and investigations.
CO4	Use qualitative research methods to conduct research
CO5	Explain the Data Collection Techniques and testing

PO – CO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	-	-	1	-	2	-	-
CO2	2	1	2	1	2	2	-	3
CO3	2	2	-	3	1	2	-	1
CO4	3	-	2	1	2	3	-	2
CO5	2	1	-	3	1	2	-	2

Note: ‘-’=Not Relevance; ‘1’=Low Relevance; ‘2’ = Medium Relevance; ‘3’=High Relevance

Course content**Theory****Unit 1****12 Hours****Research Methods in Computer Science**

Analytical vs. Empirical Methods, Theoretical vs Empirical methods, Quantitative, Qualitative, and Mixed Methods, Surveys, Case Studies, Controlled Experiments, Ethnography and Action Research, Choosing research methods, Validity threats.

Unit 2**12Hours****Empirical Research Framework:**

Logical foundations of empiricism, Scientific methods, Research Problems, Literature Reviews, Introduction to Quantitative Research, Study Designs, Controlled Experiments, Quasi-experiments, Elements and Methods, Example Experiments.

Unit 3**12 Hours****Data Collection Techniques and testing:**

Analysis and Interpretation of Quantitative Data, Descriptive Statistics, Sampling, sampling distribution, Parameter Estimation, Statistical Inference, confidence interval and Hypothesis Testing, Tests of significance, test of difference of mean and proportions, t-tests, ANOVA, Chi-square Tests, correlation and regression, Data visualization.

Unit 4**12 Hours****Qualitative Methods:**

Study Designs, Elements, and Methods, the nature and types of qualitative research, Coding Strategies, Grounded Theory, Phenomenography, Data collection methods - primary and secondary sources, Data validation, Data analysis for qualitative methods.

Unit 5**12 Hours****Other Methods:**

Survey Research, Sampling Methods, Survey Study Designs, Introduction to Mixed Methods Research, Study Designs and Method, Case studies - Single and Multi-case, Longitudinal Case Studies, Approaches to Data Collection, Intervention Methods - Action Research, Pilot Studies, Benchmarking.

Text Books

1. Research Methods in Computer Science by Arvind Kumar, 2017.
2. Empirical Research for Software Engineering: A Concise Guide to Theory and Practice by Burak Turhan, O. C. Zeynep Filiz, Ayse Bener, 1st Edition, 2016.
3. Research Methods: The Basics by Nicholas Walliman, 1st Edition, 2017.

Reference Books

1. Research Methodology in Computer Science: An Introduction by B. S. Shajitha, 1st Edition, 2018.
2. The Craft of Research, Fourth Edition by Wayne C. Booth, Gregory G. Colomb, Joseph M. Williams, 4th Edition, 2016.

Part B: Course Evaluation System

Assessment System	Assessment Component	Description	Weightage	Marks
Theory Continuous Internal Evaluation (CIE)	CIE-I	Mid Semester Examination (MSE) –I of 1 hour duration for 25 marks	--	50
	CIE-II	Mid Semester Examination (MSE) –II of 1 hour duration conducted for 25 marks	--	
	--	Average of MSE-1 and MSE-II	25	
	CIE-III	Assignments/Quizzes/Case Analysis	10	
	CIE-IV	Publications/Seminar/Quiz etc.	10	
	CIE-V	Attendance*	5	
CIE (Theory)			60%	*30M
Practical Continuous Internal Evaluation (CIE)	CIE-VI	Practical Test(s) of 30 Marks with Marks Distribution for each test as follows: 1; Write Up- 25 % 2: Execution – 35 % 3: Modification/Testing – 25% 4: Viva – 15 % Continuos Evaluation in the form of Record Write Up, Viva etc ----- 20Marks	50	
CIE (Practical)			40%	*20 M
Semester End Evaluation (SEE)	SEE	Written Examination conducted for 3 hours duration **	50%	*50 M
TOTAL MARKS			100%	100

*Attendance Marks Allotment: 90% and above : 5 Marks

85% to 89% : 4 Marks

80% to 84% : 3 Marks

75% to 79% : 2 Marks

** Semester End Examination to be conducted for 3 hours duration for Courses with 3 or more credits; 2 hours duration for Courses with 2 credits and 1 hour duration for Courses with 1 credit.

Question Paper Pattern:

The SEE question paper comprises 5 Modules/Units. Each Module/Unit will have 2 questions carrying 20 marks each. Each question may have sub questions (a, b & c etc) for a total mark of 20. Students need to answer 5 questions selecting one FULL question from each module.

**50% weightage given to SEE and 50% weightage to CIE as per Scheme of evaluate